

Review OMICs 2025

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Content

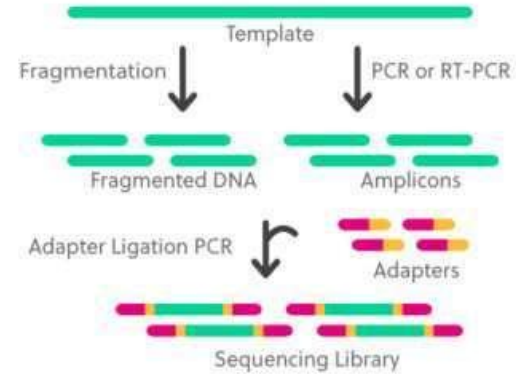
- DNA-seq
- Bulk RNA-seq
- Histone modifications
- DNA methylation
- Integration Analysis

DNA-seq

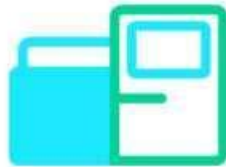
STEP 1: Extraction



STEP 2: Library Prep



STEP 3: Sequencing

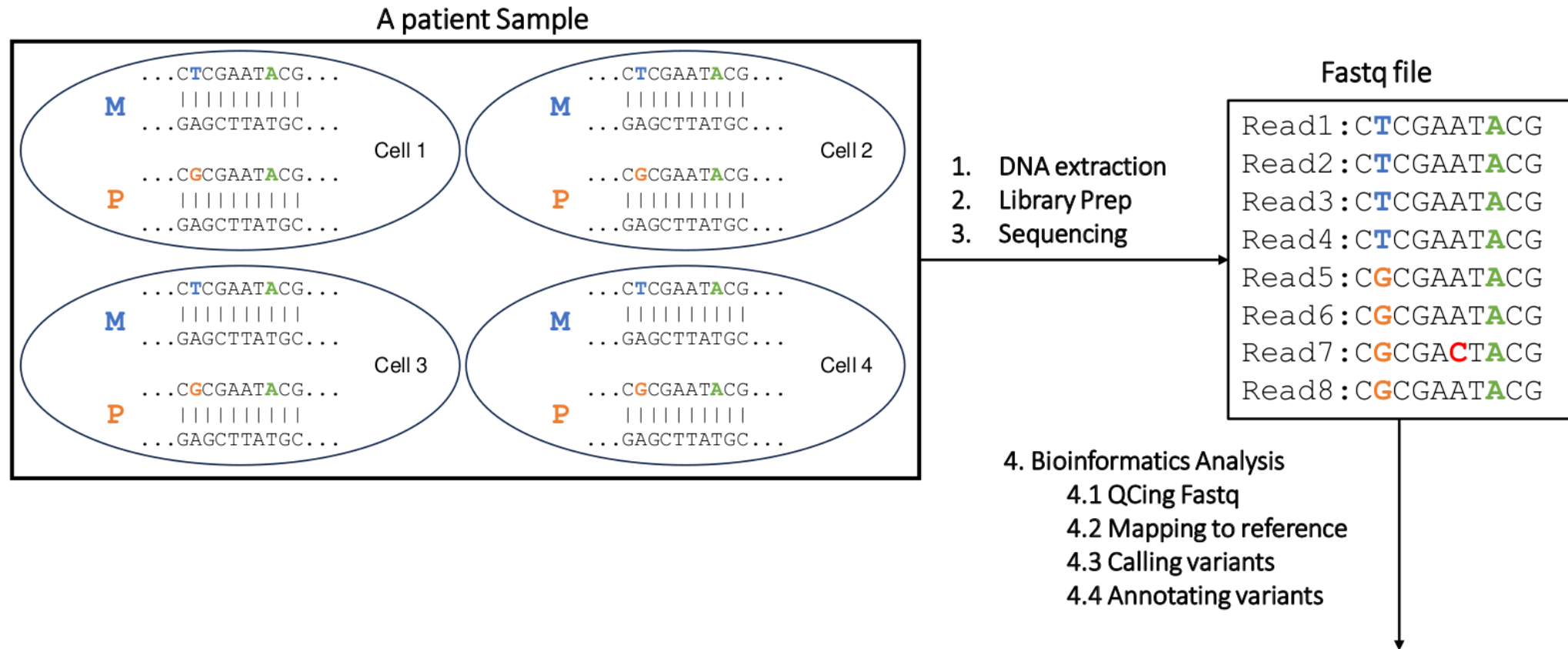


STEP 4: Analysis

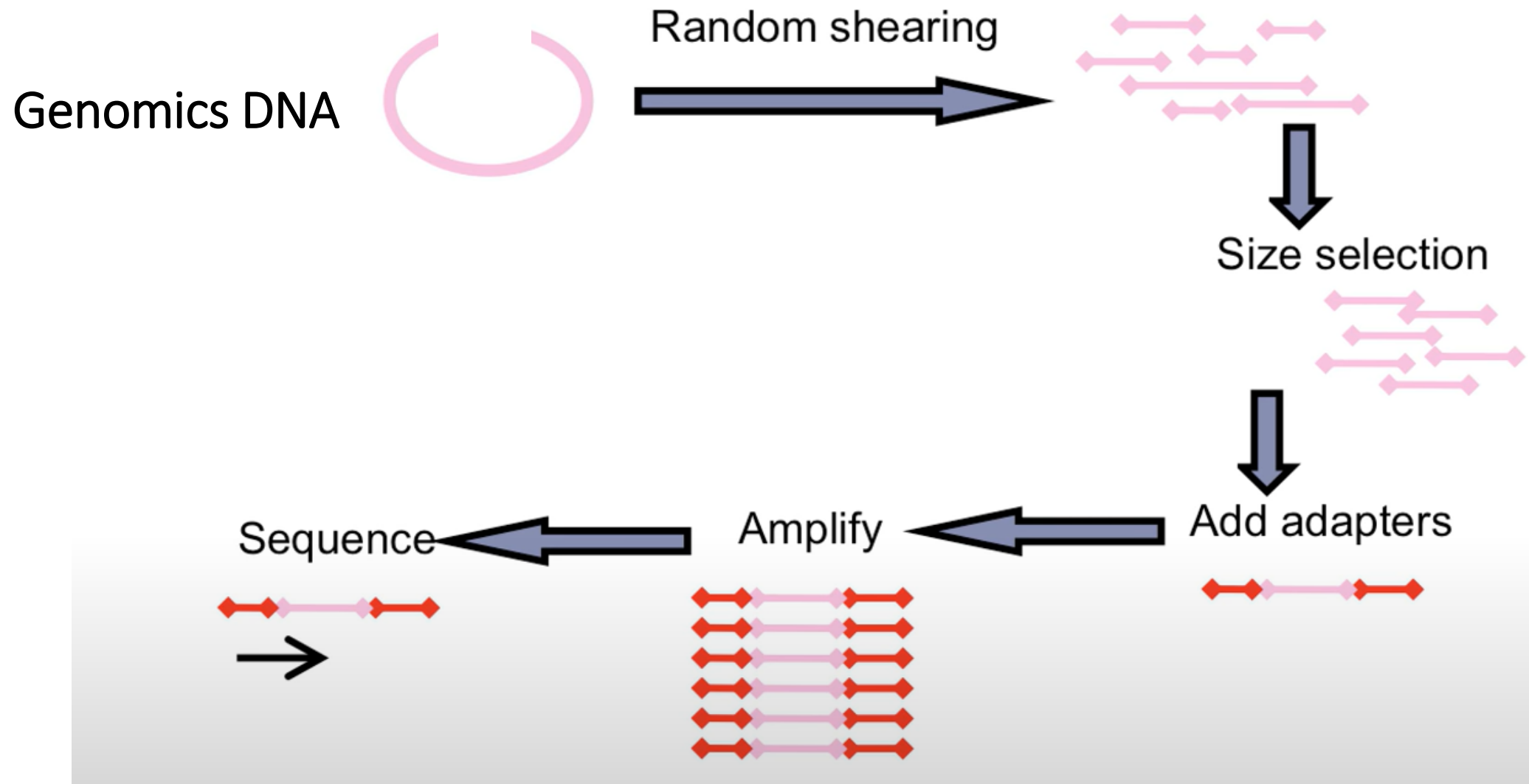


Quy trình **XÉT NGHIỆM** gen bằng
phương pháp giải trình tự thế hệ mới
(NGS)

Các bước trong XÉT NGHIỆM gen bằng phương pháp giải trình tự thế hệ mới (1)



Whole genome sequencing – Shotgun Library Prep



Các bước trong XÉT NGHIỆM gen bằng phương pháp giải trình tự thế hệ mới (2)

4.2 Mapping reads to reference

Reference: ...CTCGAATGCG...

Read1: CTCGAATACG

Read2: CTCGAATACG

Read3: CTCGAATACG

Read4: CTCGAATACG

Read5: GCGAATACG

Read6: GCGAATACG

Read7: GCGACTACG

Read8: GCGAATACG

Maternal

Paternal

Error

Heterozygous

Homozygous

4.3 Calling variants

ANN=G|stop_gained|HIGH|OR4F5|ENSG00000186092|transcript|ENST0000641515.2|protein_coding|3/3|c.822T>G|p.Trp274*|882/2618|822/981|274/326||Pathogenic

ANN=A|frameshift_variant|HIGH|ZSWIM2|ENSG00000163012|transcript|ENST00000295131.3|protein_coding|9/9|c.1238G>A|p.Ile413|1293/2451|1238/1902|413/633||;LOF=(ZSWIM2|ENSG00000163012|1|1.00)

4.4 Annotating variants

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##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=AD,Number=2,Type=Integer,Description="Read depth for each allele">

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	FORMAT	Sample1
20	14370	rs6054257	T	G	129	PASS	GT:GQ:DP:AD	0/1:48:8:4,4
20	17330	.	G	A	150	PASS	GT:GQ:DP:AD	1/1:49:8:8,8

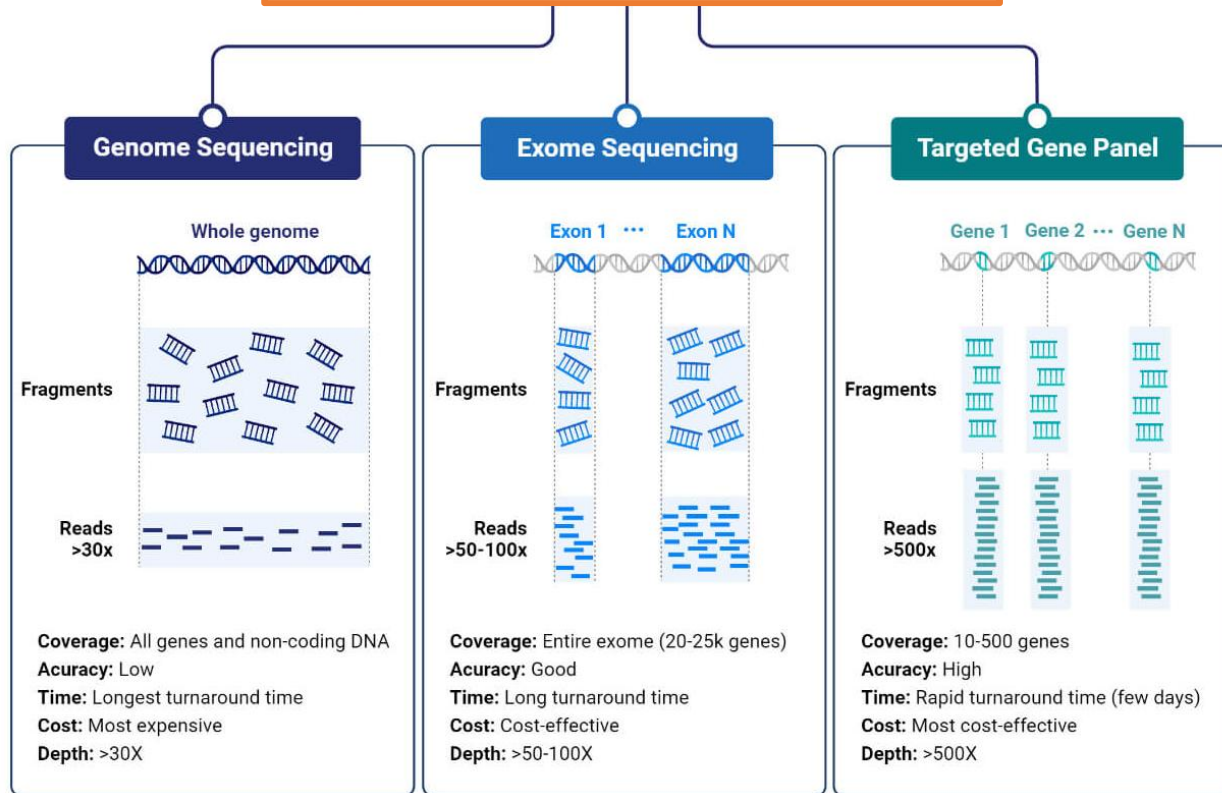
Hội đồng Hệ Gen: phiên giải và hội chẩn những biến thể trong báo cáo kết quả NGS



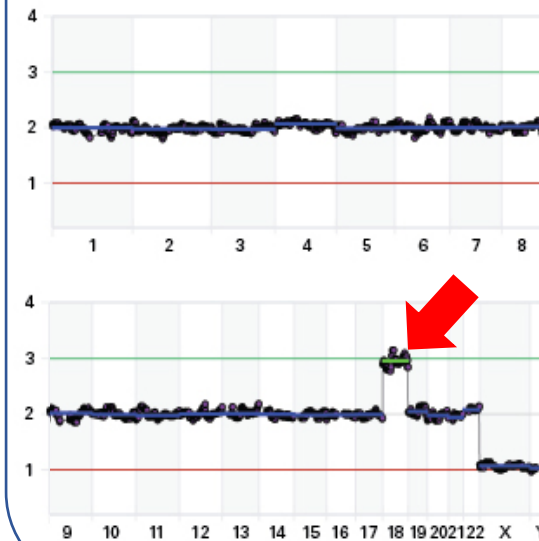
<https://ggba.swiss/en/the-first-medical-genomics-center-opens-in-geneva/>

Ứng dụng giải trình tự gen thế hệ mới trong lâm sàng

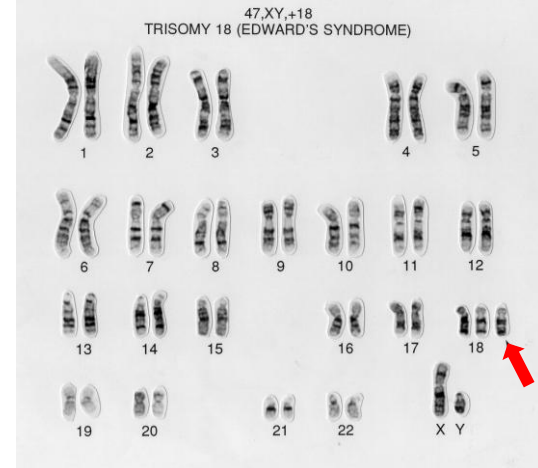
Single Nucleotide Variant (SNV)



Copy Number Variation (CNV) Low-Pass Genome Sequencing



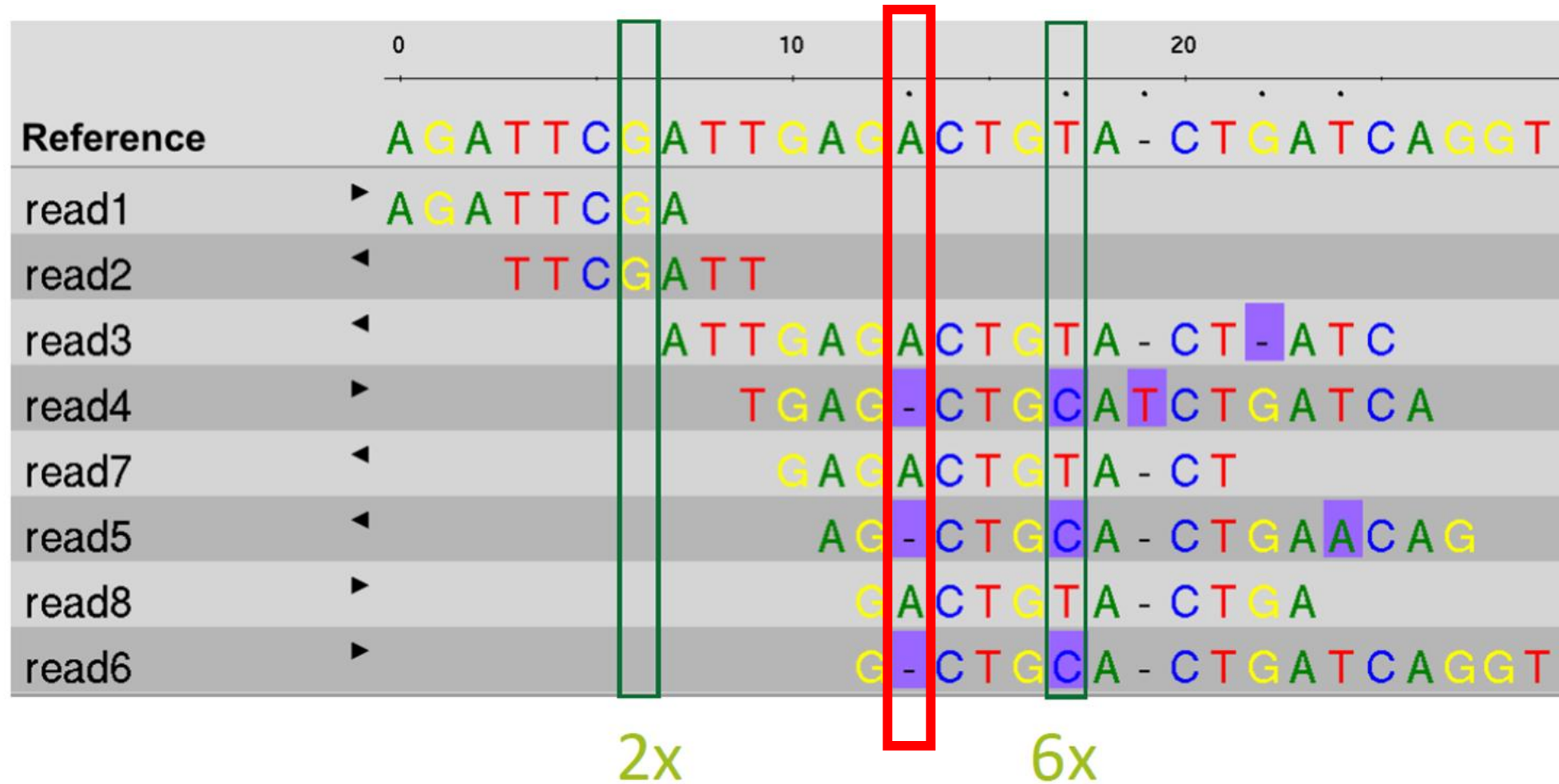
Karyotype



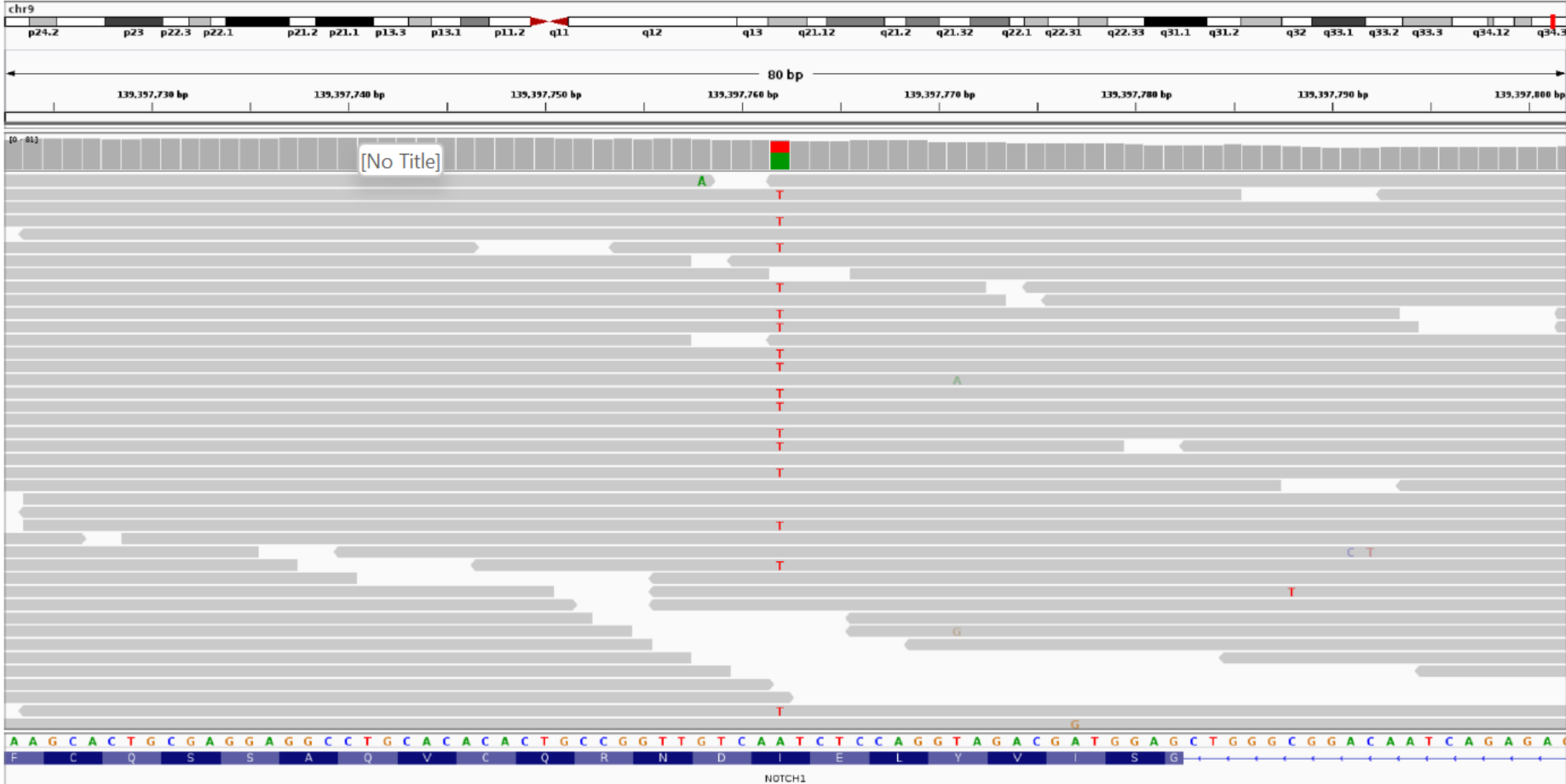
<https://microbenotes.com/next-generation-sequencing-ngs/>

<https://wellcomecollection.org/search/images?query=eaahzt2u#>

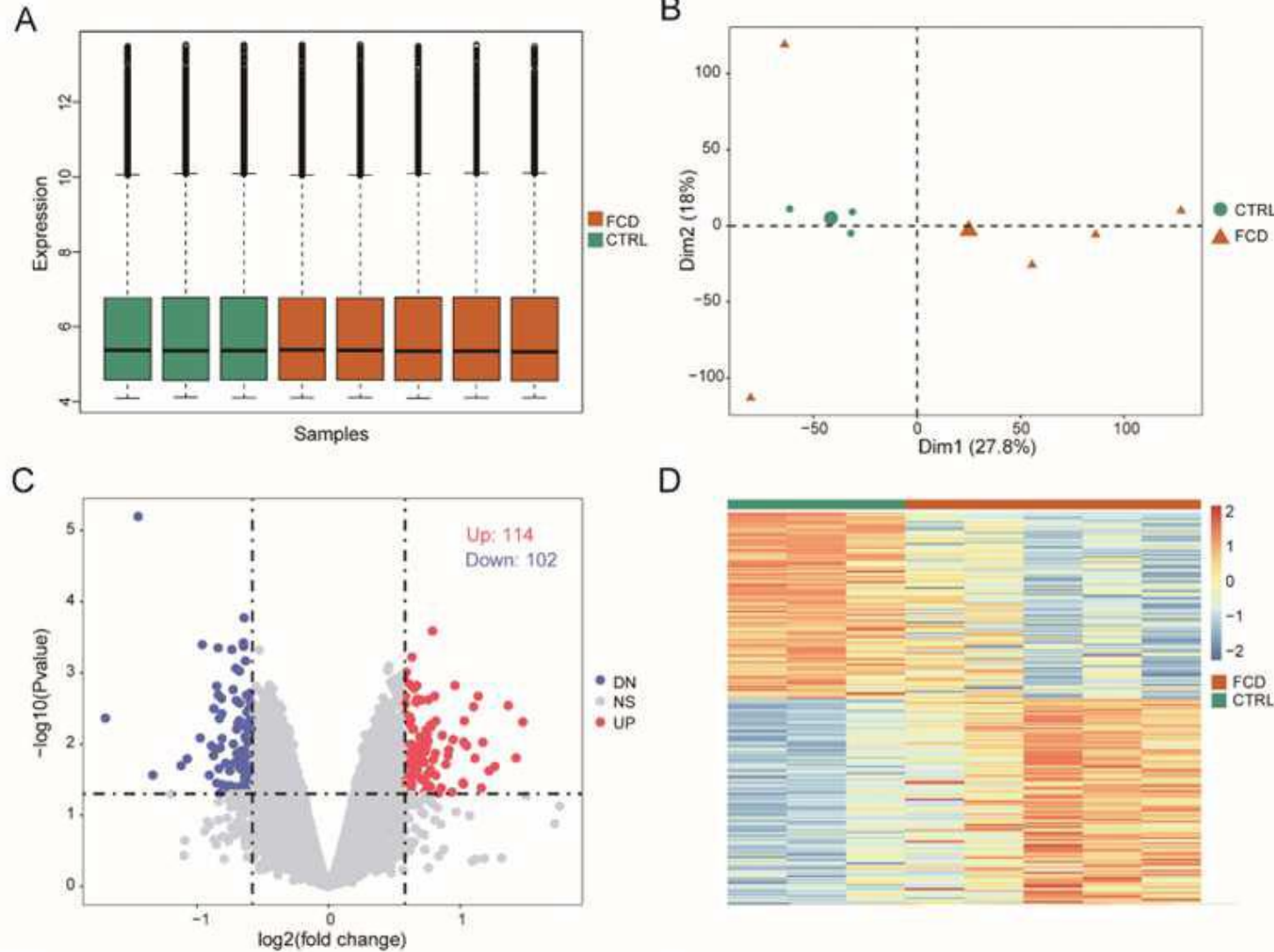
Variants in PAH



Alignment and variant viewers

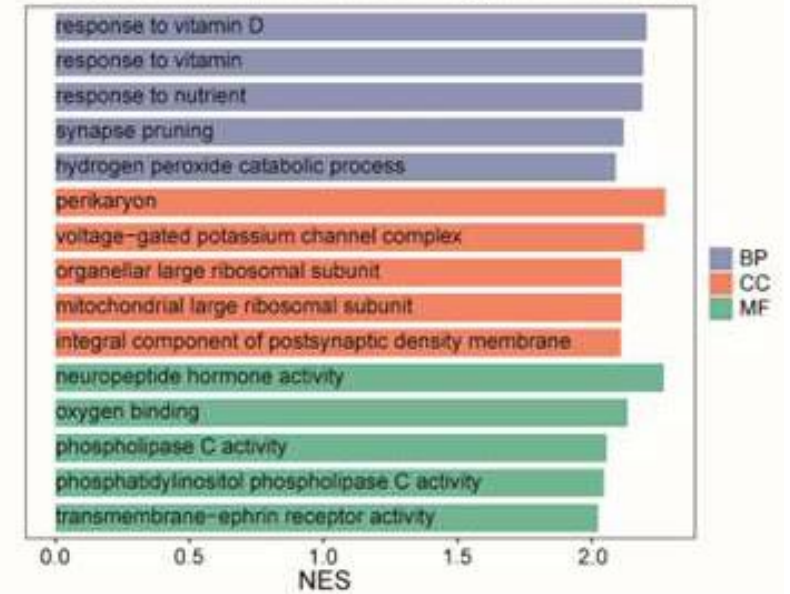


Bulk RNA-seq



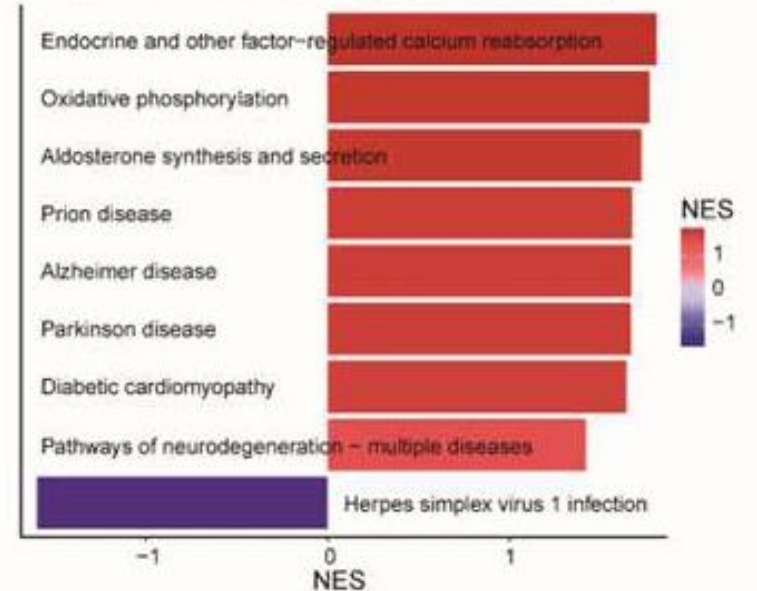
E

The TOP 5 Enriched GO Terms (FCD vs CTRL)



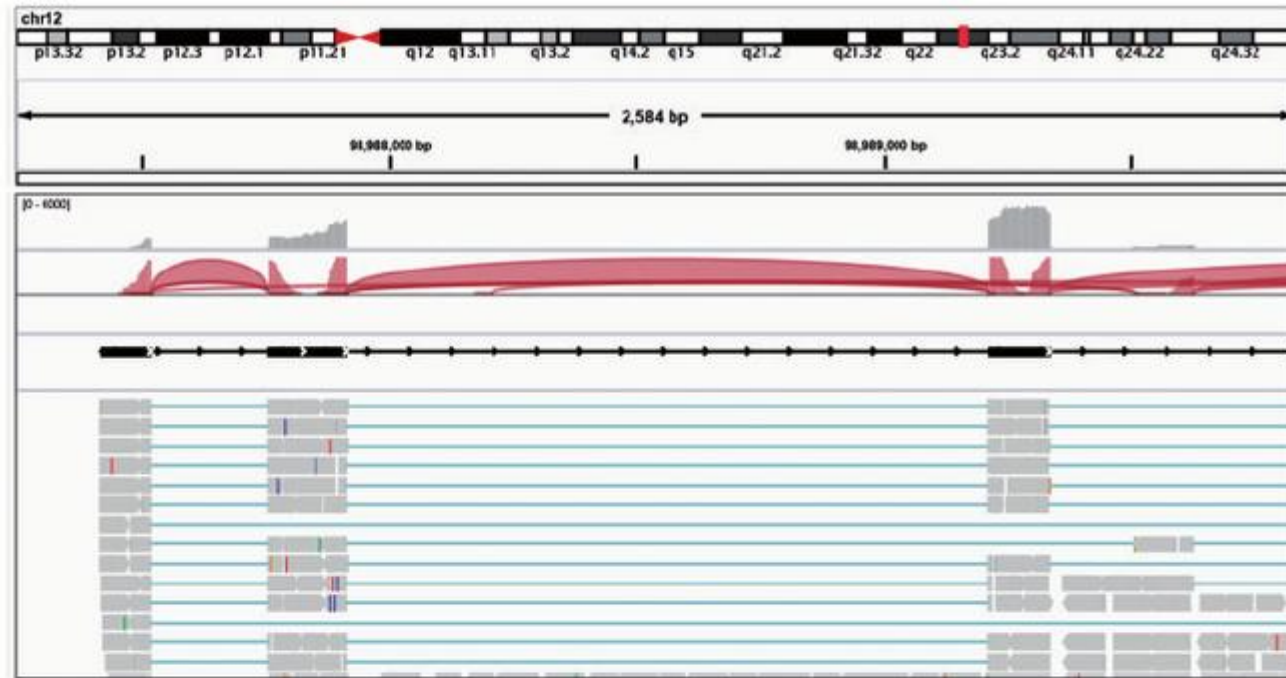
F

Enriched KEGG pathways (FCD vs CTRL)



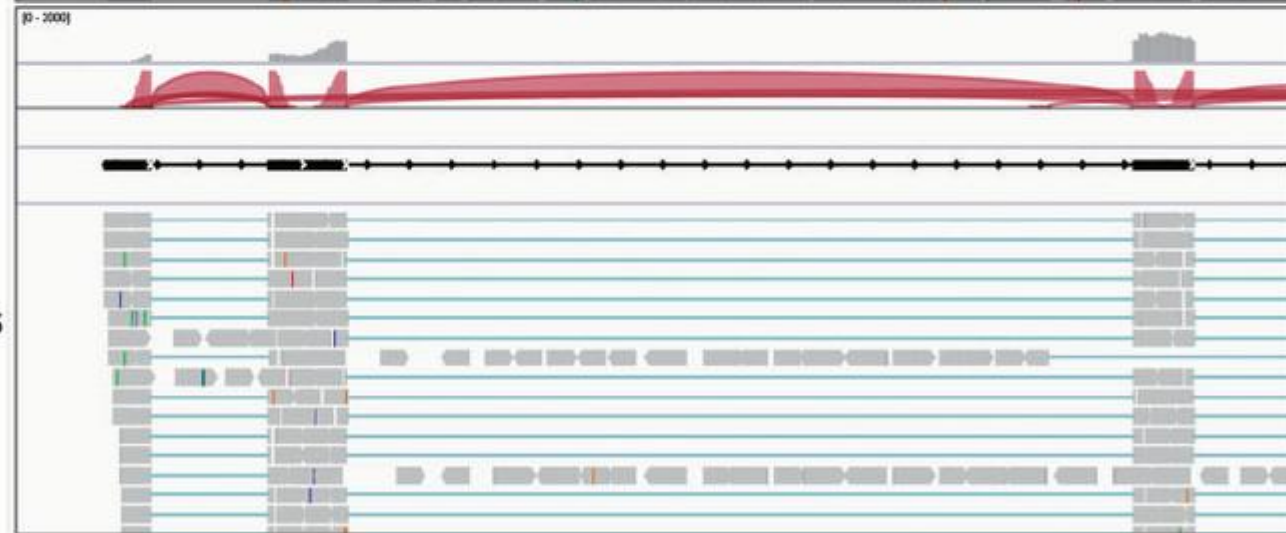
Heart

Coverage
Junctions
Transcripts
Alignments



Liver

Coverage
Junctions
Transcripts
Alignments



Refseq



Molecular interaction levels in epigenetics

4 DNA-methylation

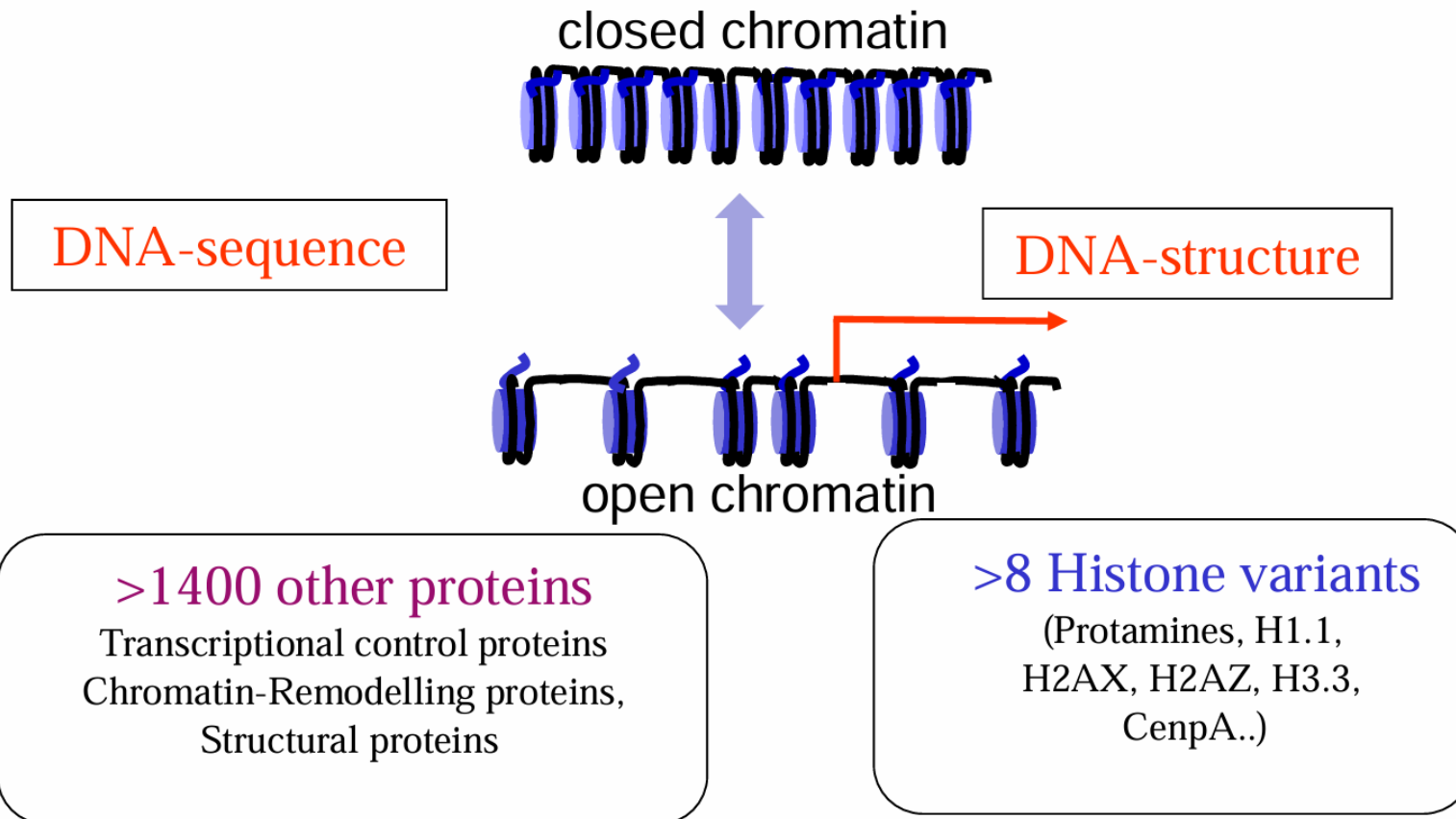
(5^{me}cytosine, 5^{hme}cytosine,
5^{carboxyl}cytosine)

> 10 classes of RNAs

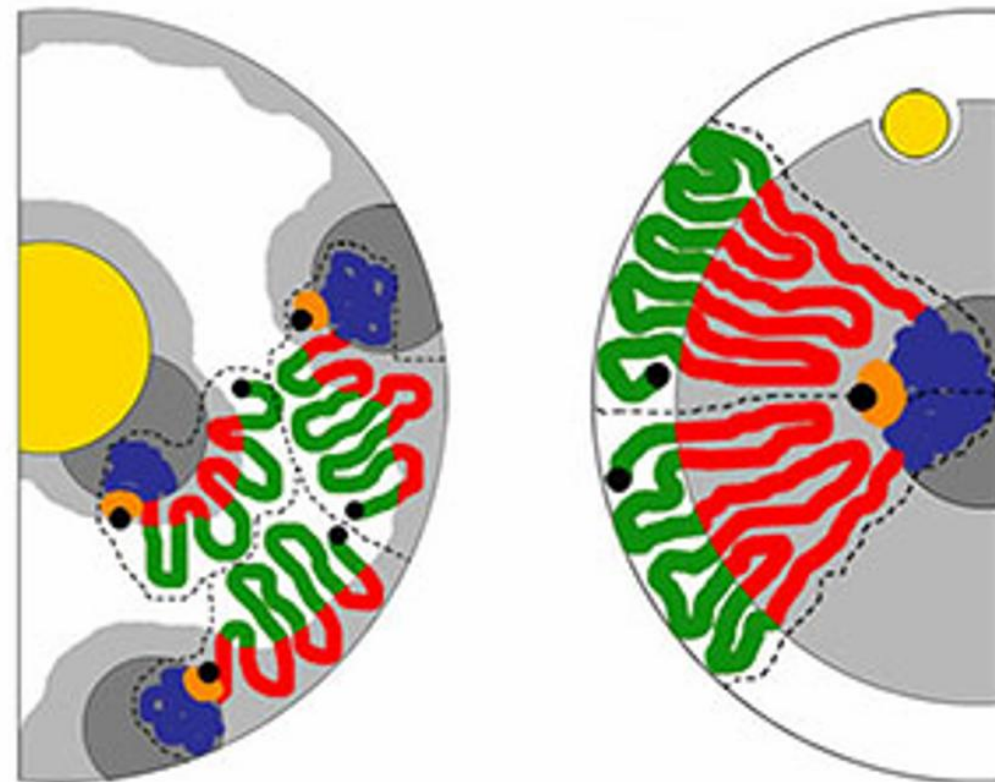
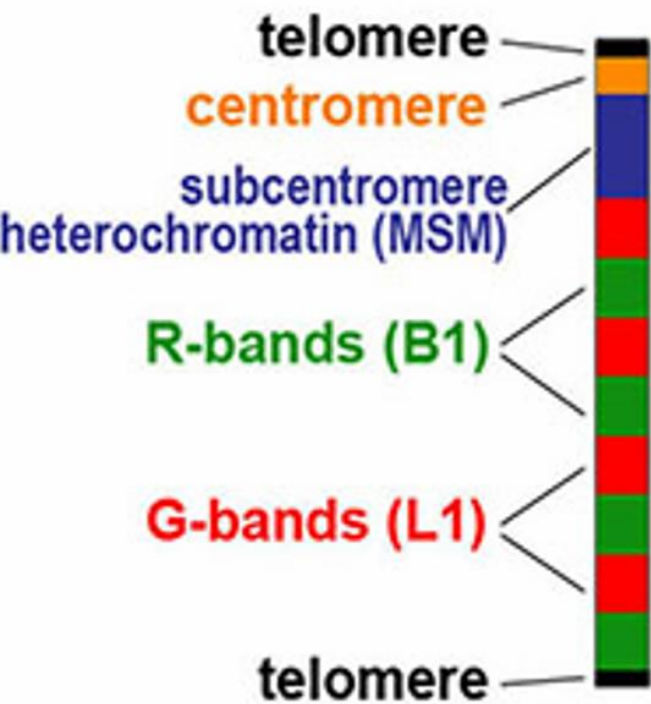
(mi-, pi-, si- and
long non-coding RNAs)

>140 Histone Mod's

(Methylation, Acetylation,
Phosphorylation,...)

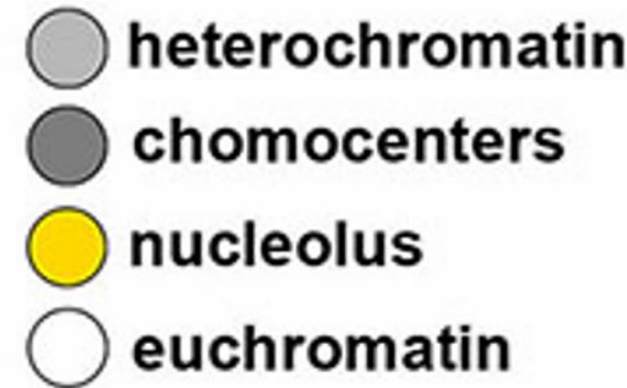


Chromatin in the nucleus: distribution of hetero- and euchromatin

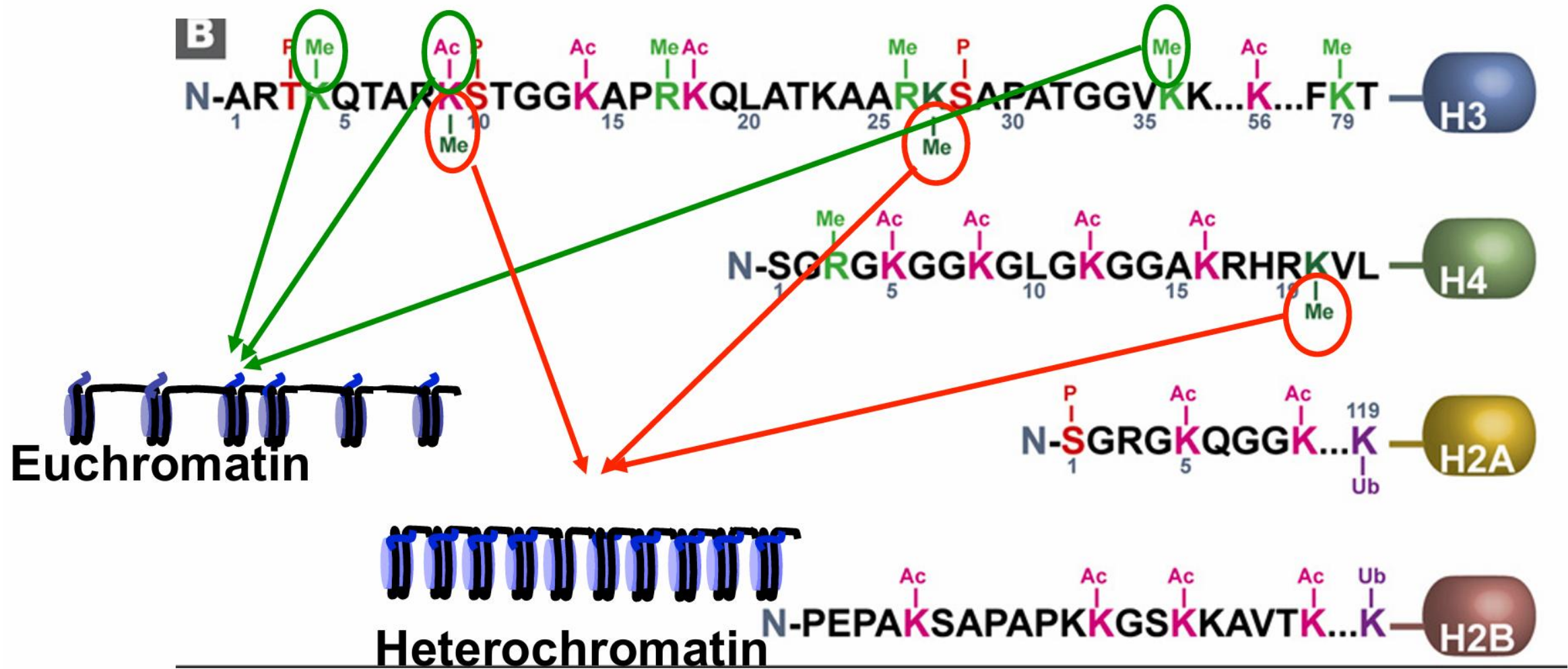


conventional nucleus

inverted nucleus



Histone methylation and open and closed chromatin structures





H3K27me3

H3K9me3

H3K4me1

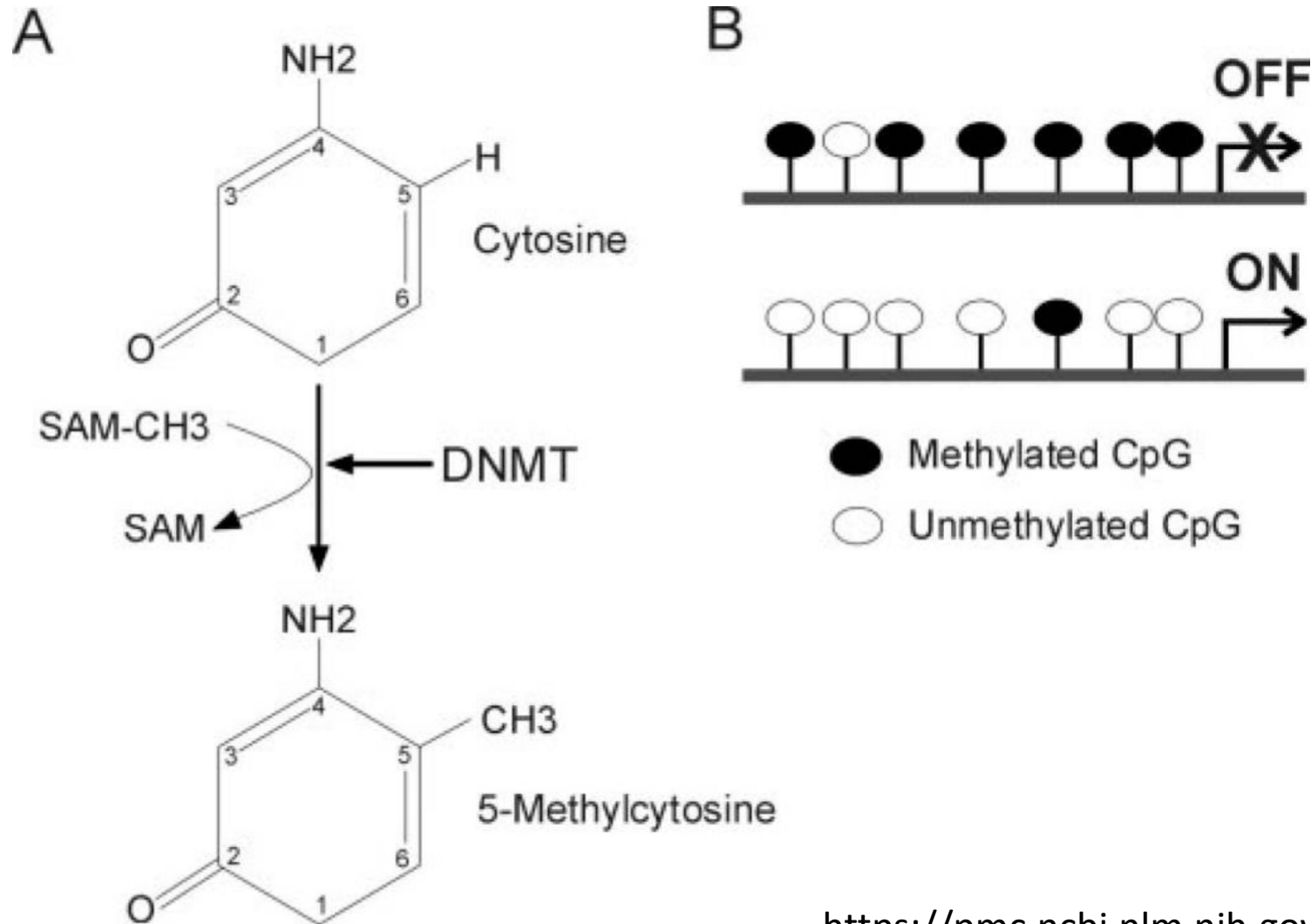
H3K4me3

H3K36me3

DNaseI open chromatin

mRNA-Seq

DNA methylation regulates gene expression





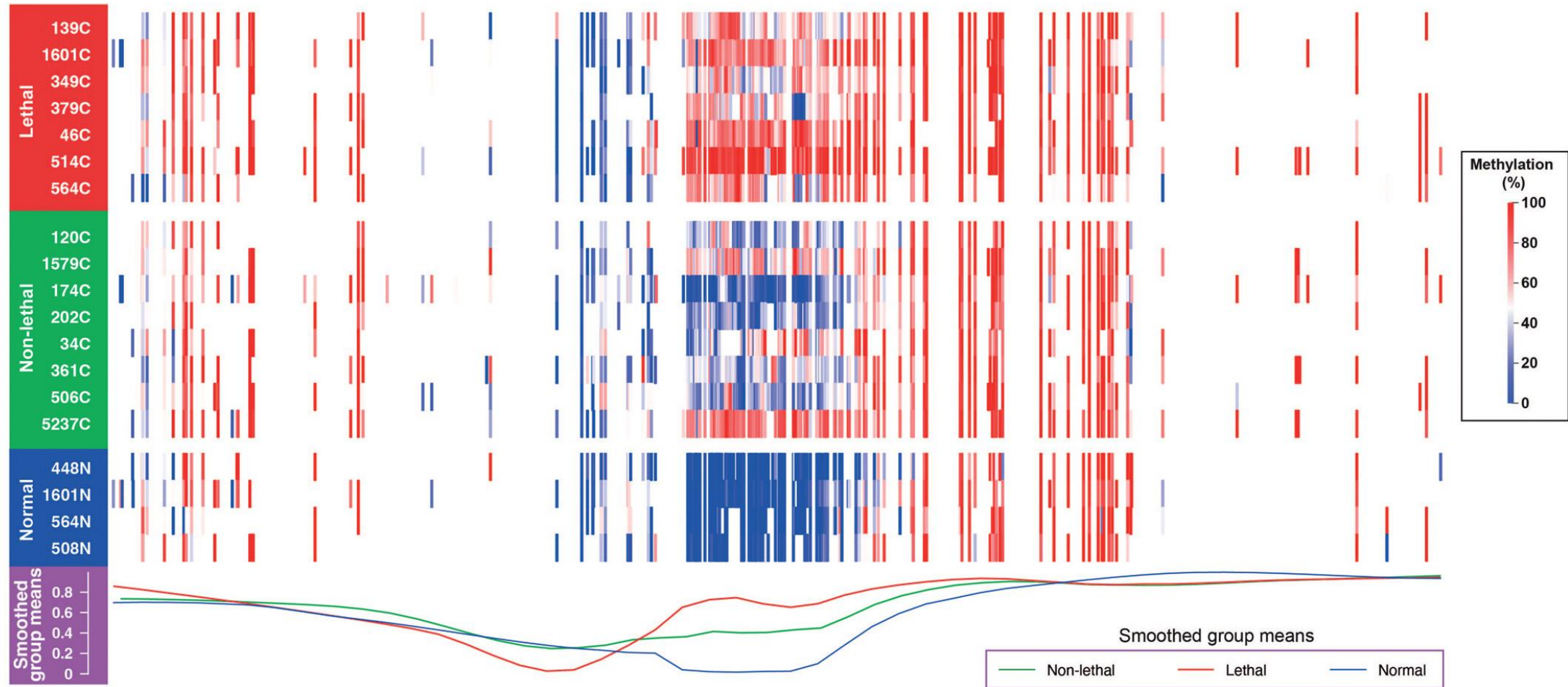
GSTP1 gene



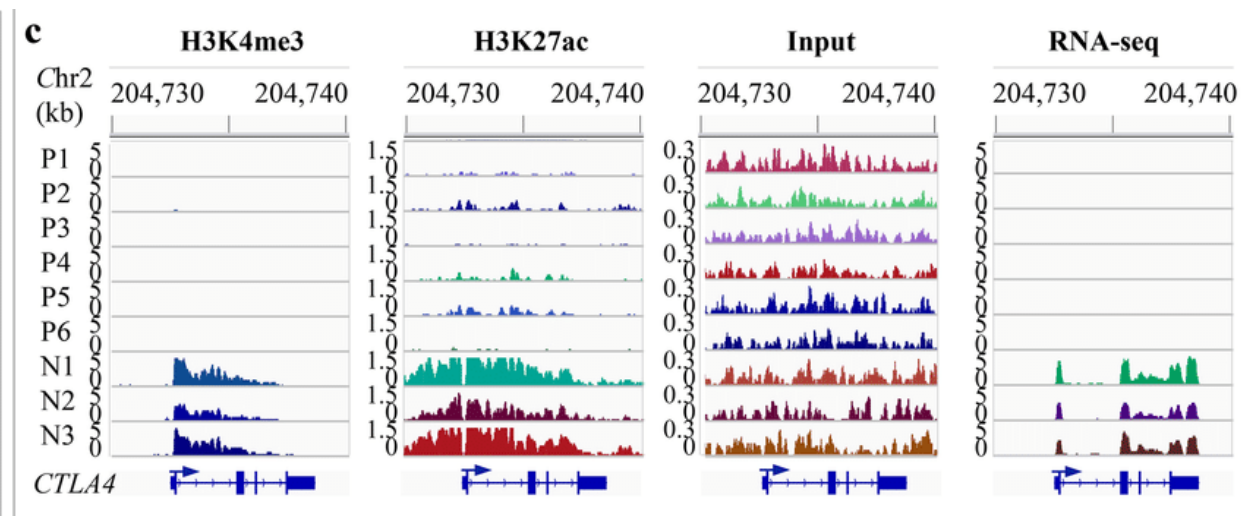
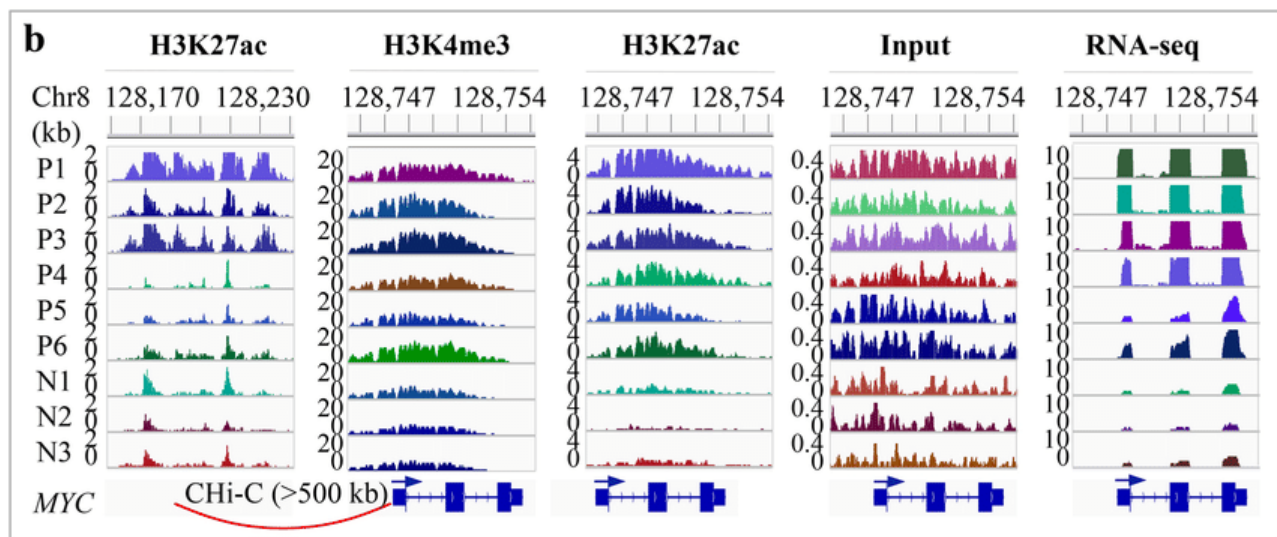
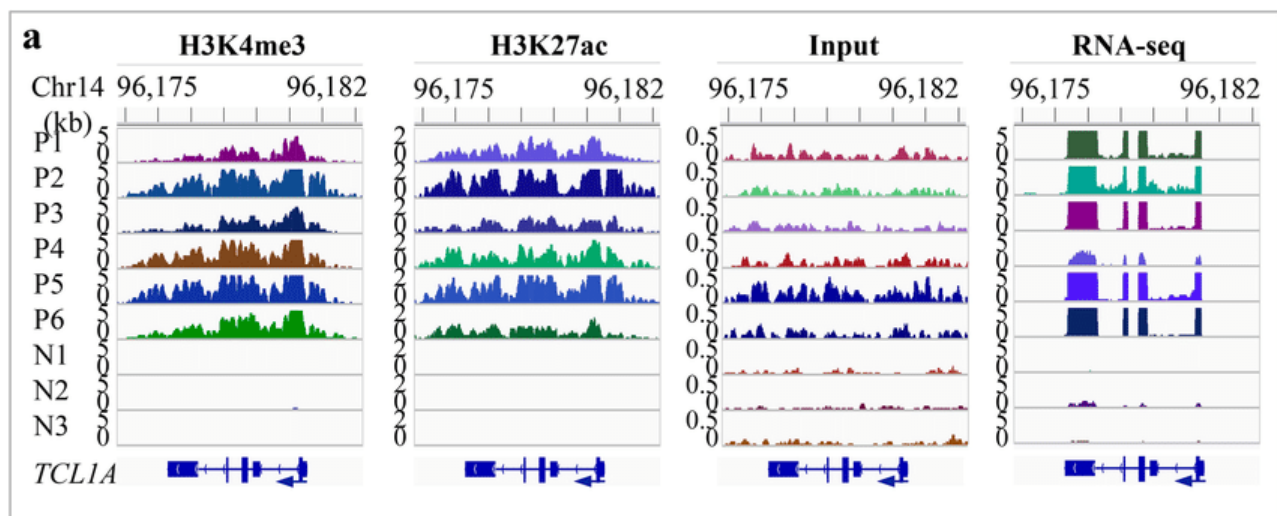
CpG island

DMR

Headloop

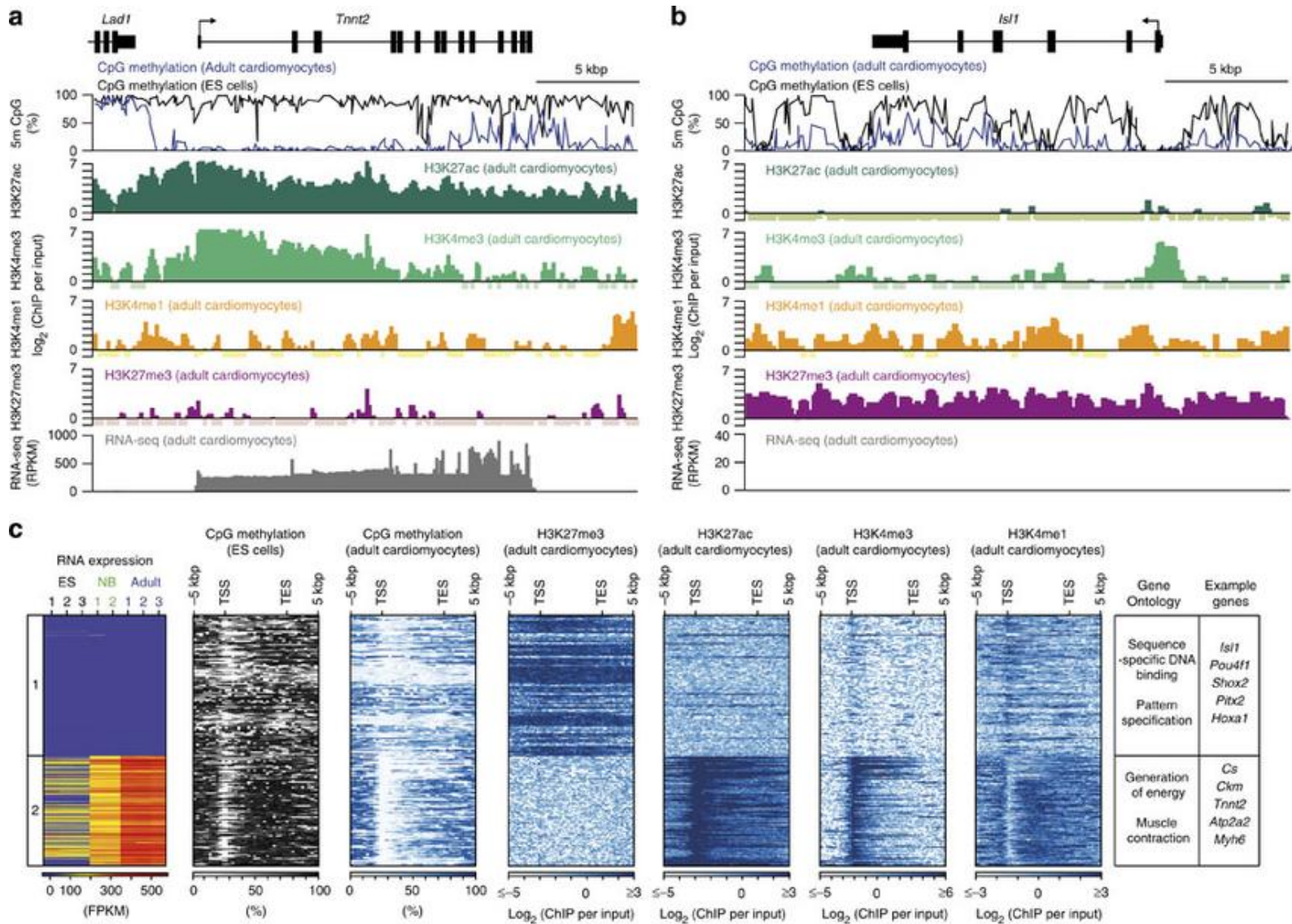


Integration Analysis

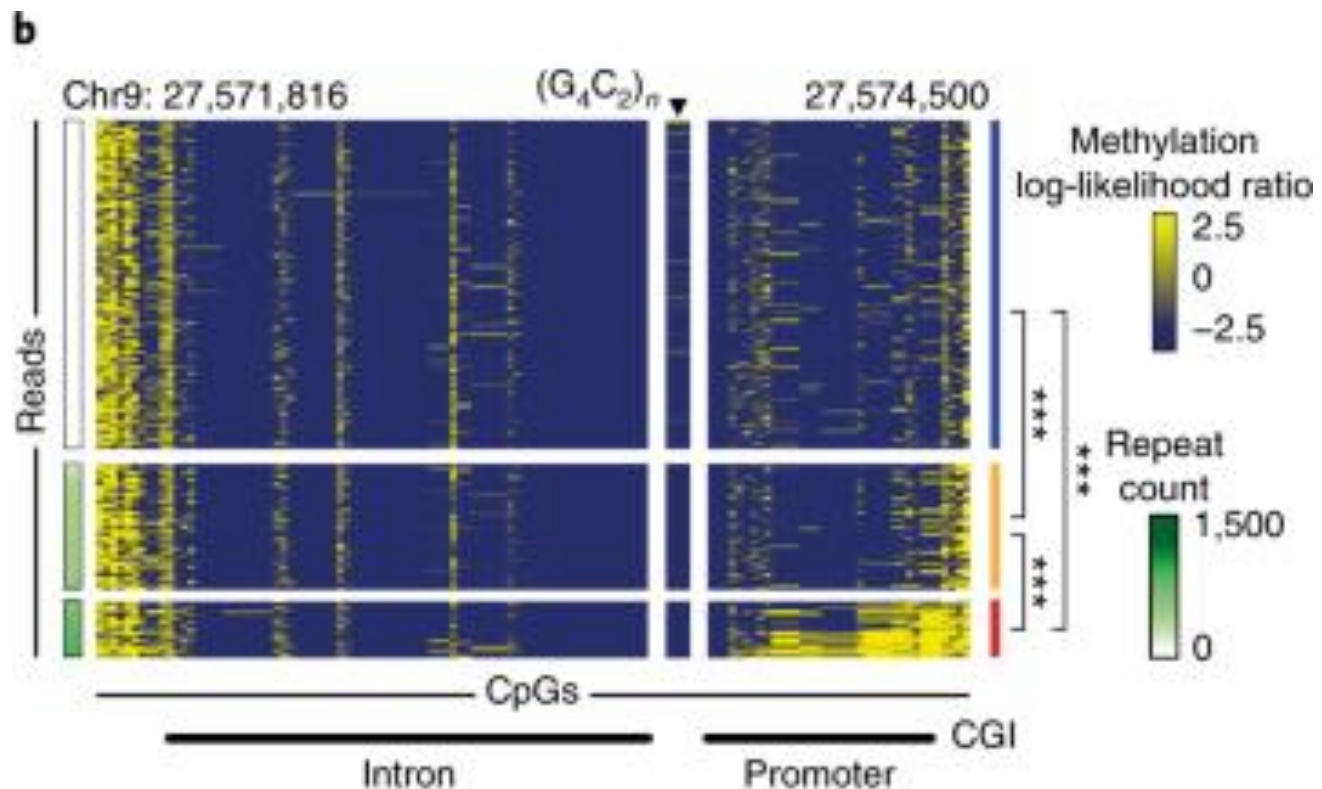
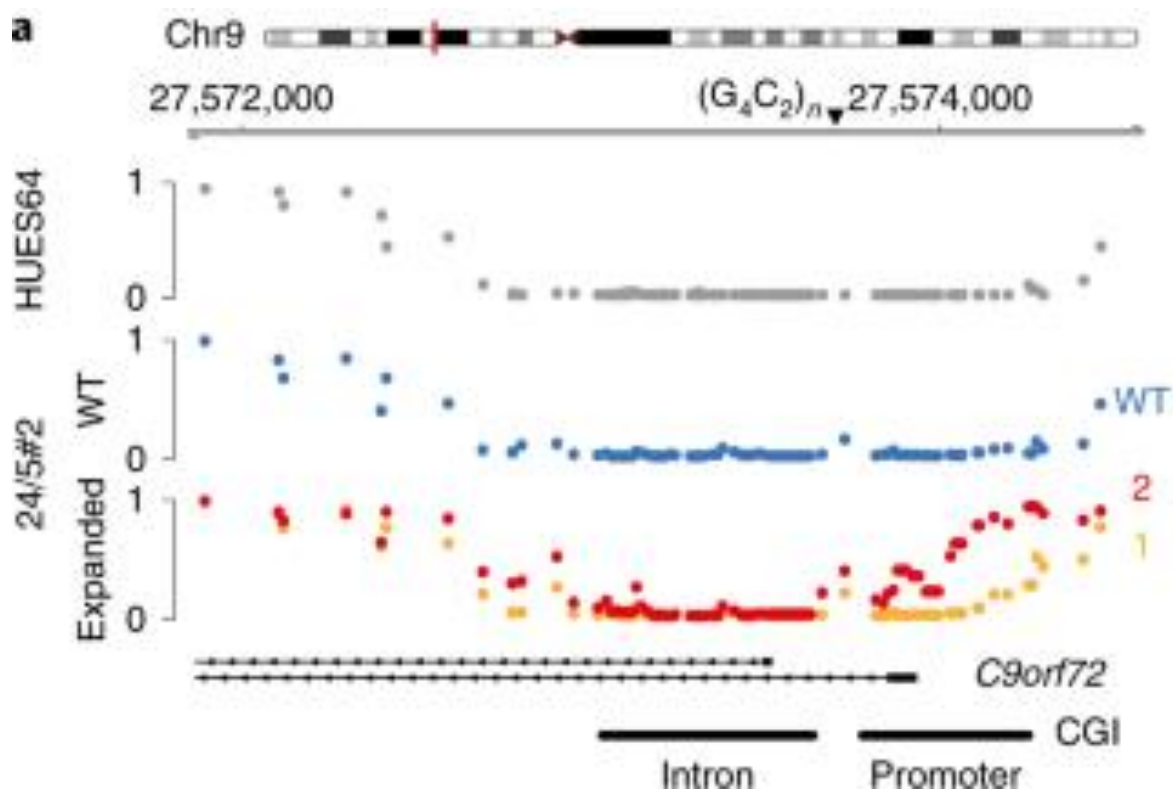


IGV snapshots showing the changes of histone marks and gene expression in T-PLL. (a) Increased expression of *TCL1A* and the associated gains of H3K4me3 and H3K27ac. *TCL1A* was actively expressed in T-PLL but not in normal. All T-PLL cases gained both H3K4me3 and H3K27ac peaks (~6-kb) that covered the promoter and gene body. Based on the read density in the inputs, there are copy number gains in two T-PLL (P1 and P5). (b) Increased expression of *MYC* and the associated increase of H3K4me3 and H3K27ac. *MYC* expression was up-regulated in T-PLL. There was an overall increase in H3K4me3 and H3K27ac occupancy over an ~6 kb region that covers the promoter and gene body. The signal level in the inputs suggested a copy number gain in P1, P5 and P6. Promoter capture Hi-C data from 17 blood cell types suggested that an ~50-kb enhancer located >500 kb upstream interacted with *MYC* promoter in total B, naïve B, fetal thymus and five types of T cells. This enhancer showed a marked increase of H3K27ac in three of the T-PLL (P1-P3) compared to the normal. (c) Markedly reduced expression of *CTLA4* and the associated loss of H3K4me3 and H3K27ac. The signal in the inputs suggested no copy number alteration within *CTLA4* in T-PLL. N1-N3, normal; P1-P6, T-PLL. Y-axis indicates the number of reads per 200-bp (for ChIP-seq and input) or per base pair (for RNA-seq).

Integration Analysis



Integration Analysis



Xin chân thành cảm ơn!